

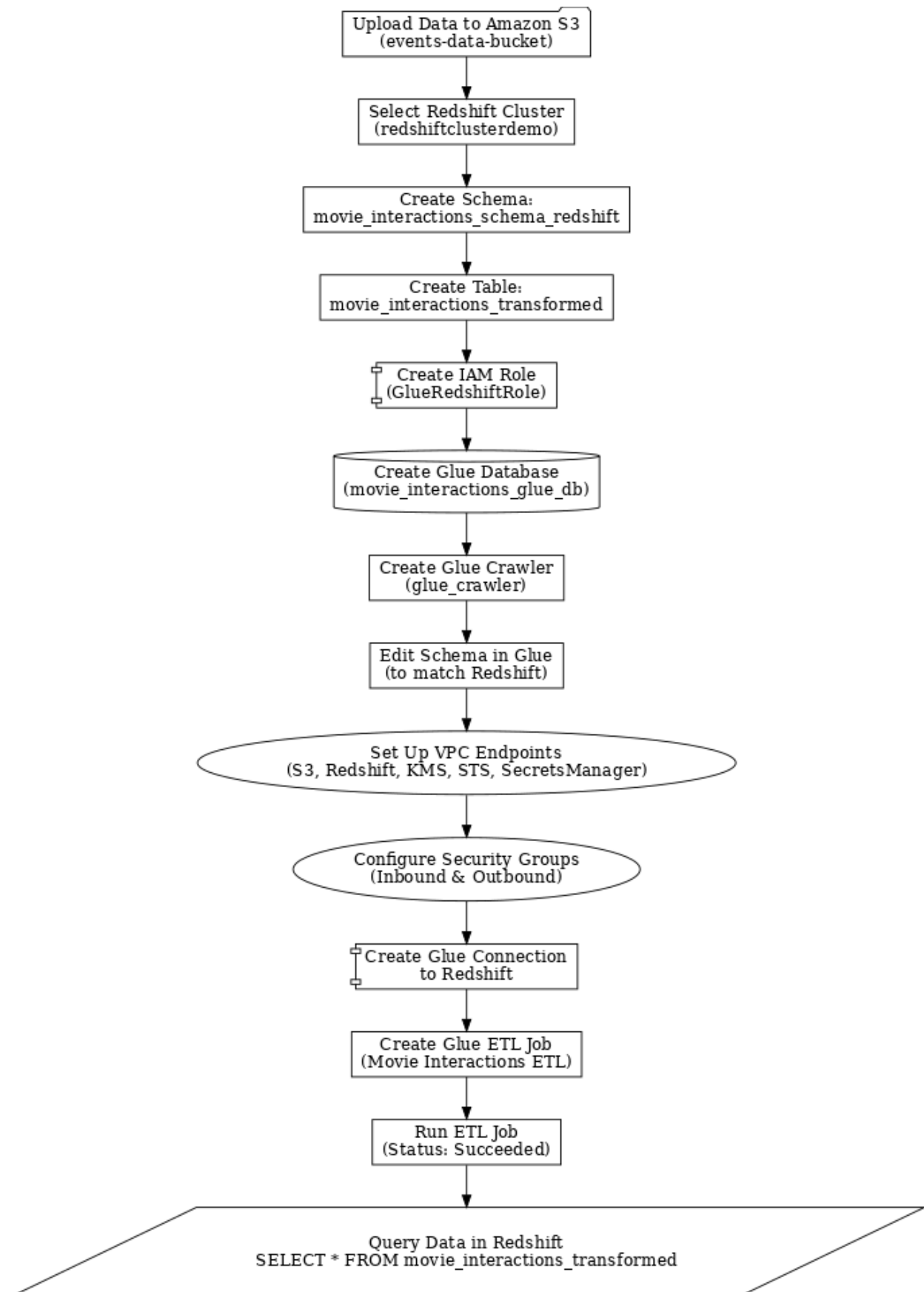


table).

So the flow is:

” Ask ChatGPT

Copy code

```
S3 (CSV) --[Crawler→Data Catalog]--> Glue ETL (transform) --[Glue connection]--> Redshift table
```

You then query the cleaned table from Redshift.

REDSHIFT:

Amazon Redshift > Clusters > Edit admin credentials

Edit admin credentials

Change admin user password

Admin password
Select an option to manage your admin password.

☐ Manage admin credentials in AWS Secrets Manager [Info](#)

☒ **Generate a password**
Amazon Redshift generates an admin password.

New admin user password

XXXXXXXXXX

☐ Show password

☐ Manually add the admin password
Manually enter the admin password.

Cancel **Save changes**

Save the password and we have to use in glue

The screenshot displays the AWS Glue console interface. On the left, a navigation pane shows various AWS Glue components, with 'Data Catalog' highlighted. The main content area is titled 'glue_crawler' and includes a 'Run crawler' button. Below this, the 'Crawler properties' section is expanded, showing a table of configuration details. The 'Data sources' tab is selected, displaying a list of data sources to be scanned by the crawler.

Name	IAM role	Database	State
glue_crawler	Glueredshiftrole	movie_interactions_glue_db	READY

Description	Security configuration	Lake Formation configuration	Table prefix
-	-	-	raw_

Maximum table threshold
-

Advanced settings

Crawler runs | **Schedule** | **Data sources** | **Classifiers** | **Tags**

Data sources (1) [Info](#) [Edit](#) [Remove](#) [Add](#)

The list of data sources to be scanned by the crawler.

Type	Data source	Parameters
Table	movie_interactions_glue_db	

Crawler properties

Name sampe	IAM role Glueredshiftrole	Database movie_interactions_glue_db	State READY
Description -	Security configuration -	Lake Formation configuration -	Table prefix raws_
Maximum table threshold -			

► Advanced settings

Crawler runs | Schedule | Data sources | Classifiers | Tags

Crawler runs (1/1) [Refresh] [Stop run] [View CloudWatch logs]

The list of crawler runs for this crawler.

Filter data
 Filter by a date and time range

🔔 **Announcing new optimization features for Apache Iceberg tables**
Optimize storage for Apache Iceberg tables with automatic snapshot retention and orphan file deletion. [Learn more](#) ↗

movie_interactions_glue_db

Last updated (UTC)
September 28, 2025 at 04:48:30  [Edit](#)

Database properties

Name	Description	Location	Created on (UTC)
movie_interactions_glue_db	-	-	September 28, 2025 at 03:5

Tables (2)

Last updated (UTC)
September 28, 2025 at 04:51:47  [Delete](#) [Add tables using crawler](#)

View and manage all available tables.

Filter tables										<
☐	Name	Database	Location	Classification	Deprecated	View data	Data quality	Color		
☐	raw_events_data_b	movie_interactions	s3://events-data-bi	CSV	-	Table data	View data quality	View		

Edit schema to match redshift schema in table in glue db

us-east-1.console.aws.amazon.com/glue/home?region=us-east-1#/v2/data-catalog/tables/edit-schema-json/

Search [Alt+S]

Tables > raws_events_data_bucket9c1c > Edit schema as JSON

```
7    "Name": "user_id",
8    "Type": "int",
9  },
10 {
11   "Name": "movie_id",
12   "Type": "int",
13 },
14 {
15   "Name": "interaction_type",
16   "Type": "string",
17 },
18 {
19   "Name": "event_timestamp",
20   "Type": "string",
21 },
22 {
23   "Name": "device_type",
24   "Type": "string",
25 },
26 {
27   "Name": "session_id",
28   "Type": "string",
29 },
30 {
31   "Name": "duration_seconds",
32   "Type": "bigint",
33 },
34 ]
```

INT

Create vpc endpoints and edit security groups:

Endpoints (6) Info

Find endpoints by attribute or tag

Name	VPC endpoint ID	Endpoint type	Status	Service
s3-gateway-endpoint	vpce-02c66b702bbbe779a	Gateway	Available	com.amazonaws.s3
redshift-interface-endpoint	vpce-0c74dec2f2e04b9c9	Interface	Available	com.amazonaws.redshift
sts-interface-endpoint	vpce-01cf5e74fc2e7cabe	Interface	Available	com.amazonaws.sts
secretsmanager-interface-endpoint	vpce-071d250cdb0936396	Interface	Available	com.amazonaws.secretsmanager

Select an endpoint

Edit inbound rules Info

Inbound rules control the incoming traffic that's allowed to reach the instance.

Inbound rules Info

Security group rule ID	Type	Protocol	Port range	Source	Description - optional
sgr-0498c251c7f6abcf	All TCP	TCP	0 - 65535	Custom	

Add rule

Cancel Preview changes

Edit outbound rules Info

Outbound rules control the outgoing traffic that's allowed to leave the instance.

Outbound rules Info

Security group rule ID	Type	Protocol	Port range	Destination	Description - optional
sgr-099e67e1ee940930f	All TCP	TCP	0 - 65535	Custom	
sgr-0cc32454a6d6a4309	HTTPS	TCP	443	Custom	

Add rule

Cancel Preview changes Save

Glue connection to connect to redshift:

10. Navigate back to **Endpoints**, and wait until all endpoints have a **Status of Available**. Click **Refresh** periodically until they do.

Create a Glue connection

1. Navigate to **AWS Glue**, toward the right side, a bit further down, click on **Go to connections**, then click on **Create connection**.
2. For the Data source, select **Amazon Redshift**, and in the bottom right click **Next**.
3. At the **Configuration connection** page, fill in the following values:
 - **Database instances:** Select **redshiftclusterdemo**.
 - **Password:** Paste in the password that you stored earlier.
 - **IAM service role:** Select the **GlueRedshiftRole**.
4. Click **Next**, then click **Next** again.
5. Click **Create connection**.

Use password here

The screenshot shows the 'Configure connection' page in the AWS Glue console. On the left, a sidebar lists navigation options: 'Connectors', 'Data sources', 'Jobs', 'Crawlers', 'Databases', 'Tables', 'Views', 'Partitions', 'Queries', 'Dashboards', 'Alerts', 'Connections', 'Endpoints', 'Catalog', 'Data quality', 'Data security', 'Data lake', 'Data science', 'Data engineering', 'Data integration', 'Data migration', 'Data replication', 'Data retention', 'Data transformation', 'Data validation', 'Data visualization', 'Data warehouse', 'Data analytics', 'Data science', 'Data engineering', 'Data integration', 'Data migration', 'Data replication', 'Data retention', 'Data transformation', 'Data validation', 'Data visualization', 'Data warehouse', 'Data analytics'. The main content area is titled 'Configure connection' and contains the following fields:

- Database instances:** A dropdown menu showing 'redshiftclusterdemo-ccnc3dm3d3sy' with a refresh icon.
- Database name:** A text input field containing 'interactions_db'.
- Credential type:** Two radio buttons: 'Username and password' (selected) and 'AWS Secrets Manager'.
- Username:** A text input field containing 'admin'.
- Password:** A text input field with masked characters (dots).
- Enforce SSL - optional:** An unchecked checkbox.
- IAM service role:** A dropdown menu showing 'GlueRedshiftRole' with a refresh icon.

At the bottom right, there are 'Cancel' and 'Previous' buttons.

aws [Search] [Alt+S] United States (N. Virginia)

mviet

Last modified on 9/28/2025, 12:25:46 AM Actions

Visual Script Job details Runs Data quality Schedules Version Control

+ Data source - Data Catalog AWS Glue Data Catalog

Data preview (16) info READY

Filter sample dataset

interaction_id	user_id	movie_id	interaction_type	event_timestamp	device_type
1	101	1001	click	2025-02-02 10:00:00	mobile
1	101	1001	click	2025-02-02 10:00:00	mobile
2	102	1002	click	2025-02-02 11:00:00	mobile

Data source properties - Data Catalog

Name: AWS Glue Data Catalog

Database: Choose a database. movie_interactions_glue_db

Table: raw_events_data_bucket9c1c

Partition predicate - optional: Enter a boolean expression supported by Spark SQL, using only partition columns. Partition predicate syntax for Spark SQL is year == year[date_sub(current_date, 7)] AND month[date_sub(current_date, 7)] AND day == day[date_sub(current_date, 7)].

And load data to redshift from glue

aws [Search] [Alt+S] United States (N. Virginia)

et2

Job has not been saved Actions

Visual Script Job details Runs Data quality Schedules Version Control

+ Data target - Amazon Redshift

Redshift connection

Choose the AWS Glue connection from Amazon Redshift, or create a new connection

Redshift connection

Database: interactions_db

Schema: Choose your Amazon Redshift schema. movie_interactions_schema_redshift

Table: Search and enter the name of the source Amazon Redshift table. movie_interactions_transformed

Handling of data and target table

☒ APPEND (insert) to target table: AWS Glue will append data to existing columns of the table and discard any extra columns.

☐ MERGE data into target table: AWS Glue will either update or append data to the table based on a set of conditions.

☐ TRUNCATE target table: Same as Append, except AWS Glue will first clear the contents of the table.

☐ DROP and recreate target table: AWS Glue will delete and recreate the table with the schema from the source data.

Target node not supported: You have selected a data target node which is not supported for data preview. Please select another type of node instead.

In redshift:

us-east-1.console.aws.amazon.com/redshiftv2/home?region=us-east-1#query-editor:

Search [Alt+S] United States (N. Virginia) Account ID: 6290-1243-0765 cloud_user

Run Save Schedule Clear Send feedback

Query results Table details

Query 4185 [Execution](#) [Data](#) [Visualize](#)

Completed, started on September 28, 2025 at 01:02:29
ELAPSED TIME: 00 m 02 s

Rows returned (10) [Export](#)

Search rows

interaction_id	user_id	movie_id	type	event_time	device_type	session_id	duration_seconds
1	101	1001	click		mobile	session_001	120
2	102	1002	view		tablet	session_002	300
3	101	1001	click		mobile	session_003	180
4	103	1003	view		desktop	session_004	150
5	104	1004	click		mobile	session_005	220
6	105	1005	view		tablet	session_006	250
7	106	1006	click		desktop	session_007	200
8	107	1007	view		mobile	session_008	310
9	108	1008	click		tablet	session_009	130

2:10 9/28/25